

Atmospheric Simulation

Simulation Team

28 September 2026

A Brief++ feature tour

Example Research Lab

Abstract

A small, self-contained report showing the document model and all seven output formats.

1 Introduction and scope

This report combines **measurements** with an *idealized* model. The word **Brief++** refers to the report builder. See the project source for the code and 1 for the generated figure. The report builder is documented by [1].

A report should preserve the meaning of its content across formats.

note: The numbers below are illustrative rather than observational data.

2 Model

The pressure model uses a scale height $H = 8.4$ km under a constant-temperature assumption.

$$\frac{dP}{dz} = -\rho g \quad (1)$$

$$P(z) = P_0 e^{-z/H} \quad (2)$$

Equation 2 describes the synthetic curve used here.

```
double pressure(double z, double p0, double h) {  
    return p0 * std::exp(-z / h);  
}
```

3 Results

Table 1: Illustrative sea-level conditions

Parameter	Value	Unit
Temperature	273.15	K
Pressure	101325	Pa
Scale height	8.4	km

The values in 1 produce the trend shown in 1.

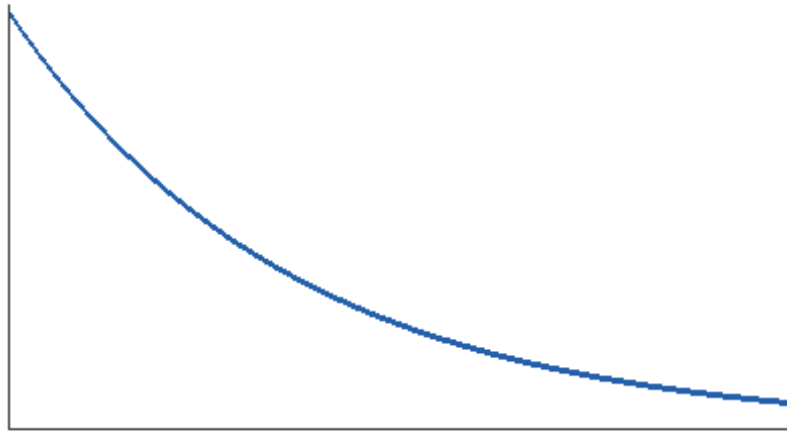


Figure 1: Synthetic atmospheric **density** by altitude.

1. Set the starting conditions
2. Review the output in every format
3. Compare the rendered documents

- **Check the details**

- Check the figure and table
- Check the linked references

Density Mass per unit volume

Scale height Altitude over which pressure falls by a factor of e .

3.1 Interpretation

The example emphasizes report structure, not atmospheric accuracy.

warning: Do not use these illustrative values for a real forecast.

tip: Edit this source file and rebuild the demo to compare formats.

Table 2: Headerless summary table

Input	Illustrative profile
Outputs	Seven text formats and a PDF

This paragraph was added from a `DocumentFragment` to demonstrate reusable content.

4 Backend-specific notes

LaTeX-only note.

References

- [1] Brief++ contributors. Brief++ project source and documentation. 2026. <https://github.com/vrtulka23/briefpp>